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AI/ML intern

KEY LEARNINGS REPORT

Introduction

The Raritone Summer Internship 2026 provided an opportunity to explore the application of Artificial Intelligence and Computer Vision in the fashion industry. Throughout the internship, our team researched various technologies related to body detection, pose estimation, virtual try-on systems, fashion recommendation systems, and AR/VR-based shopping experiences. This internship helped us gain both technical knowledge and professional skills while working on a collaborative project.

Technical Learnings

1. Human Body Detection and Pose Estimation

One of the primary areas explored during the internship was human body detection and pose estimation. We learned how AI models can identify body landmarks and track human movements using computer vision techniques. Through research on MediaPipe and OpenPose, we gained an understanding of how body keypoints can be extracted and used for applications such as virtual try-on systems, body measurement estimation, and motion tracking.

2. Body Measurement and Landmark Analysis

The internship introduced us to the concept of estimating body measurements from images using pose landmarks. We studied how body proportions, shoulder width, waist estimation, and height estimation can be derived from detected body points. Understanding body measurements highlighted their importance in improving clothing fit and personalization within virtual try-on systems.

3. Cloth Segmentation and Human Parsing

We explored cloth segmentation techniques used to separate garments from the background and identify different clothing regions. Models such as U2Net and Detectron2 were studied to understand image masking, garment extraction, and human parsing. This research provided insights into how segmentation contributes to realistic virtual try-on experiences.

4. 2D Virtual Try-On Systems

A significant portion of the internship focused on understanding 2D virtual try-on workflows. We learned how user images are processed through pose detection, segmentation, garment alignment, and garment warping to generate virtual try-on outputs. This helped us understand the complete pipeline involved in creating AI-powered fashion fitting experiences.

5. 3D Virtual Try-On and AR/VR Technologies

We also explored advanced technologies related to 3D virtual try-on systems. Research included 3D avatar generation, body mesh reconstruction, cloth simulation, and motion tracking. We gained knowledge about tools such as Blender, Unity, and Three.js, which are commonly used in developing immersive virtual fashion experiences.

6. AI-Based Fashion Recommendation Systems

The internship provided exposure to recommendation systems used in fashion platforms. We studied content-based filtering, collaborative filtering, and hybrid recommendation approaches. We learned how AI can analyze user preferences, shopping behavior, and fashion trends to provide personalized outfit suggestions.

7. Dataset Collection and Analysis

Another important learning experience was understanding the role of datasets in AI model development. We researched datasets such as DeepFashion, DeepFashion2, and VITON-HD, which are widely used for fashion recognition, segmentation, and virtual try-on research. This helped us appreciate the importance of high-quality data in training and evaluating AI systems.

Professional Learnings

1. Team Collaboration

Working as part of a team improved our ability to communicate ideas, share knowledge, and collaborate effectively. The project encouraged teamwork and coordination while exploring different AI modules.

2. Research and Documentation

The internship strengthened our research skills by requiring us to study multiple AI models, technologies, and datasets. We also learned the importance of maintaining clear documentation and presenting findings in a structured manner.

3. Problem Solving and Critical Thinking

Analyzing different AI approaches and comparing their advantages and limitations improved our analytical and problem-solving abilities. We learned how to evaluate technologies and identify suitable solutions for specific use cases.

4. Project Coordination and Time Management

The internship helped us develop project management skills by planning tasks, tracking progress, meeting deadlines, and organizing project deliverables effectively.

Conclusion

Overall, the Raritone Summer Internship 2026 was a valuable learning experience that enhanced our understanding of Artificial Intelligence, Computer Vision, and Fashion Technology. The internship provided practical exposure to modern AI applications and helped develop both technical expertise and professional skills. The knowledge gained through this project will be beneficial for future academic projects and career opportunities in the fields of AI, Machine Learning, and Computer Vision.